Full name of the applicant	Pranav Shailesh Pandya
Reference e-space	40011852

SCIENTIFIC SECTION OF THE PROPOSAL

MAIN LANGUAGE CHOSEN = ENGLISH

This part includes the following elements:

1. Report on past research (optional)
2. Description of the research project
3. Comments on changes made in the research project in case of resubmission (optional)
4. Potential interdisciplinary approach of the research project (optional)
5. Description of the work environment
6. Summary of the master's thesis or equivalent
7. Additionnal comments (optional)

The applicant must fill in the sections below and convert the file into an unprotected PDF before appending it to the online application form.

The F.R.S.-FNRS insists on **strict compliance with the instructions given for each part of the proposal** (scientific section relevant to the instrument selected, number of pages allowed for documents to be enclosed with the application form…) and stresses again the sovereign consideration of the Scientific Commissions assessing the application file.

1. REPORT ON PAST RESEARCH (OPTIONAL)

I graduated with distinction (program topper) from Symbiosis International University with a MTech (Master of Technology) degree in Geoinformatics & Surveying Technology. I successfully completed my master thesis under the TREES (Training and Research in Earth Eco-System) program organized by Space Application Center (SAC), Indian Space Research Organization (ISRO), under the supervision of Dr. Gaurav Jain. My thesis was focused on determining optimal solar power plant locations for supplementing electrical vehicle charging using remote sensing inputs and environmental parameters. We identified a network of EV charging stations that met the governmental regulations, and we developed a balancing algorithm to better estimate the area of solar panels needed to supply energy demands while optimizing the use of barren/empty lands in the vicinity. The final output of this research methodology was the ability to develop a map containing the suitable optimized charging network that incorporates the placement solar panels in identified barren lands, while considering the optimization of solar radiant energy received by surfaces through Global Horizontal Irradiance (GHI) parameter. This GHI data was confidential and was made available due to special nature of TREES internship, making this research truly unique within the organization only, and making an important new addition to the state of art.

In my final year, I was hired as a Junior Research Fellow in Department of Science & Technology (DST) & IC-IMPACT's Indo Canada Building Energy Optimization under landscape characteristics project, which involved the use of a LoRa sensor network for hyperlocal sensing of thermal comfort across different land surfaces in different configurations. Working in this project gave me opportunity to use LoRaWAN technology to setup in-situ sensors that collected data for modeling urban thermal comfort, which shares similarities and demonstrated my expertise for the present doctoral project. After acquiring the hyperlocal data, satellite-based observations were used for ground validation of Building Energy models. Additionally, I was exposed to work with the integrating IoT sensors on Amazon Web Services (AWS). In this project we submitted a paper in Urban Climate Journal (Q1) which is under review, and presented a paper in Flame Conference on Climate Change which was held in Dec 2021, that will be published in conference proceedings in 2022.

In my senior year, my team won Global Nasa Space Apps Challenge due to our crowdsource data fusion application for critical disaster assessment, which is currently under communication with the WALMI center in Gujarat, India. This project involved the use of crowdsourced data from volunteers in flood regions, where they can take pictures using an app that applied an auto geotag to photos and added them to the SQL Database, which we could retrieve and apply a time-space comparison allowing visualization over pre and post-flood scenarios. Authorities could use this app to get a live visualization of pre & post-flood scenarios, and reduce disaster rescue action delays due to lack of information. This project involved the use of near real-time data under cloud-free and image quality filters.

My undergrad final year project received an AICTE fund of ₹1.67 million and facilitated purchasing Ideaforge's Ninja Nano UAV with Night Vision surveillance, live human tracking, and time-based crime surveillance. This project was my final year thesis in B. Tech work on Smart Crime Management in Smart Cities, which used past data scrapping from news articles and geocoding their locations in space and time to understand crime patterns, predict crime possibilities, evaluate crime vulnerability, and subsidize those vulnerabilities by adding patrolling routes and placing more police stations and patrol vehicles in such neighborhoods.

In 2021 during my final year of M. Tech, we received approval from AICTE and got the Ideaforge's Ninja Nano UAV with 3 payloads (thermal Infrared, Photogrammetry and TCC video imagery) delivered to the Birla Vishwakarma Mahavidyalaya (B. Tech) university, and a personal was authorized to conduct training for the drone flight by manufacturer. I received that training in December 2021, and each member of my team got a chance to perform VTOL, orientation and create mission plan in Ideaforge's software, and to independently deploy drones.

In December 2019, I presented two papers in Symbiosis National Conference SYMGI-2019. During this time I met with Prof. Sharvari Shukla who is the director of Symbiosis Statistical Institute (SSI), and she gave me an opportunity to contribute a Springer Book Chapter, on Geointelligence. During the Covid19 crisis, the use of Public Health, Mass Safety and Geography became important, demanding an intelligence hub containing information from land features and geophysical properties to human altercations, mobility, and resource utilization. In the Covid19 crisis, I got involved in drafting paper on the application of Geointelligence for healthcare, especially in a developing country such as India. This book is published this year in 2022.

While working on this springer book chapter, I got the change to collaborate with Prof. Prasad Pathak, who is Research Dean at Flame University. He had received the DST-ICIMPACT grant for Indo-Canada Building Energy Optimization project, and he allowed me to begin working in this project as a Junior Research Fellow, during my final year of M. Tech since April 2021. Within the bounds of JRF at Flame University, I also got the opportunity to order experimental sensors and camera lens such as fish eye lenses and a thermal imager, which were designed for gathering spectral imagery across inter-building distances to analyze the radial extent to which a surface warms its neighborhood within Urban Heat Island Effect. The use of a fish eye lens facilitated gathering 130-degree images which were useful in gathering Sky View Factor (SVF) values by orienting the camera towards Zenith and taking pictures of sky, and later digitizing the surrounding features that would limit sky view. This was also useful in linking the inbound solar radiation (Global Horizontal Irradiance) values towards the open to sky surface areas that receive solar energy for major part of day, especially when the sun is at maximum inclination. This research explored the understanding of radiation interactivity and cross-heating which warmed the air temperature registered in LoRa sensors across different surface nature & configurations.

In 2021, towards my final year of M. Tech I participated with a team of students in Mapathon competition organized by Indian Institute of Technology, Bombay, for which my team became the Mapathon Champion by competing again 9300 entries, where we had designed an Electrical Vehicle Charging Network Plan based on excess surplus energy generated at power stations, residential energy usage, and the proximity of nearby charging infrastructure using Resourcesat Satellite's LISS-III data and the algorithm of Maximum Likelihood Classifier. Petrol Stations within the range of Power Lines (to avoid transmission loss) were identified for the Electrical Vehicle Charging Network, and the areas selected where those areas requested by the Ministry of Power (2018) (https://www.youtube.com/watch?v=ubGp76LAhmE).

In final semester of my M. Tech, I won Varanasi Smart Hackathon. With this event government of Varanasi was planning to get Smart City status and they were inviting students with tech ideas for innovation to build smart solutions for many industrial problems such as transportation, pollution, etc. My team's idea was to subsidize renewable energy and use electric mobility technology. The network planning was done using satellite data and spatial city attributes taken from Open Streets Map. Existing electric infrastructure with the existing residential mapping was done for identifying the supportive locations where charging station can be established. This idea facilitated us in winning 1st place and a reward of iPad Air. The government is presently deciding their funding priorities, but they requested us to assist them in developing EV Infrastructure using our optimization solution.

I have been interested in immersive reality since 2016, when VR technology became accessible to public, with YouTube launching the first 360 immersive videos and with the release of the app VR box. I received a student license for the VR Max software by Autodesk and Open-Source software called Blender in order to learn how 3D movies can be made for VR. Further, I did Certified Training Courses on Campus during my undergrad on Quantum GIS (QGIS), IDRISI, DGPS (Differential GPS), AutoCAD, and 3D Laser Scanning. I also enrolled in Personality Development Program (PDP) organized by Globarena Pvt. Ltd. In 2017.

2. DESCRIPTION OF THE RESEARCH PROJECT

Urban GIST - Green Infrastructure for Stress reduction and Thermal comfort:

2.1 Goals of the research

The project focuses on the exploration of the role Urban Green Infrastructure (UGI) design plays in citizen well-being, with a focus on Thermal Comfort (TC) and Stress Level Reduction (SLR). While the positive role of nature on general well-being is widely recognized, the UGI implementation characteristics that most effectively contribute to the improvement of TC and SLR are yet to be identified. To address this research gap, this thesis will implement a novel pedestrian-centered research methodology combining environmental and biometric sensing, eye-tracking and immersive media technologies, and participant surveys, through the deployment of off-site and on-site experimental campaigns.

Prior literature that has focused on 'macro' or city scale studies, have highlighted the beneficial role of Nature-based Solutions (NbS) for heat island, noise or air pollution mitigation (Johnson et al. 2020; Llaguno-Munitxa and Bou-Zeid 2020a; Manoli et al. 2019). Making use of hybrid physics and data-driven models, urban fabric, microclimate and air quality, health reports, and socio-demographic metrics have been studied, and the positive role of NbS for urban health have also been reported (Sowińska-Świerkosz et al. 2021). At the 'micro' or neighborhood scale, local sensing approaches using carry-on sensing kits have enabled the study of the local environmental gradients to high spatiotemporal resolutions (Apte et al. 2017; Pigliautile and Pisello 2018) and have reported the benefits of local UGI to lower air temperature and enhancement of thermal comfort. Publications focusing on the study of the Physiological Equivalent Temperature (PET) - the complete heat budget of the human body- have combined biometric and environmental sensing approaches aiming to obtain quantifiable metrics of urban thermal comfort (Villadiego and Dabat 2014; Piselli et al. 2018). However, despite the extensive literature, research focused on i) the study the role small to medium scale UGI plays in thermal and psychological stress, and the ii) study of local UGI characteristics that are most effective to enhance thermal comfort and SLR are yet to be developed.

With the aim to address these research gaps, this thesis will focus on the development of a pedestrian-centered research methodology where objective metrics on the UGI infrastructure quantification (computed using datasets collected through LiDAR scanning, remote sensing and GIS databases), urban microclimate data (collected through a carry on sensing kit containing sensors for temperature, relative humidity, and solar radiation amongst others), and biometric data (collected through immersive media, eye-tracking, and wearable sensing technologies), will be combined with subjective metrics acquired through pedestrian survey campaigns. That is, a pedestrian-centered research methodology is proposed whereby not only environmental and physiological metrics, but also psychological aspects will be evaluated to identify small to medium scale UGI and their characteristics that most effectively contribute towards thermal comfort and stress level reduction.

The thesis will address the following research questions:

i) What is the role small to medium scale urban GI implementations and their characteristics - ratio, geometry, type, forest biodiversity- play in thermal comfort and stress level reduction?
ii) Can we through the deployment of a pedestrian-centered research methodology better report the pedestrian level multi-domain perception associated to thermal comfort?
iii) Are environmental stressors and especially so thermal stress a main driver of psychological stress?

2.2 State of the art

One third of adult's report suffering from high physiological stress levels, and these numbers are escalating, driven by global trends towards urbanization and increased socioeconomic and environmental pressures such as those recently induced by the COVID-19 pandemic (Cole et al. 2020; Sharifi and Khavarian-Garmsir 2020). Similarly, the incidence of reported urban heat stress cases is expected to worsen with global warming and urban densification, leading to synergies between Urban Heat Island (UHI) and heat waves which can cause mortality and severe heat-related illnesses. UHI phenomena is caused by the urban morphology and impervious materials -with thermal properties different to those of green areas- which lead to the modification of the energy budget and the evapotranspiration potential affecting both air UHI and surface UHI and causing thermal discomfort (Manoli et al. 2019; Founda and Santamouris 2017). Furthermore, environmental stressors such as increased temperatures have also been identified as critical drivers of psychological stress (Tawatsupa et al. 2012). The beneficial role of UGI for citizen comfort is undeniable. The possibilities of nature-based restoration approaches for SLR are already being discussed (Orchard and Schiel 2021), and UGI implementations have already proved to be very beneficial to lower air temperatures up to 6°C (Jayasooriya et al. 2017; Imran et al. 2019). However, prior studies have not yet focused on the study of the role small to medium scale UGI implementations and their spatial characteristics, also called 'nature quality', play in pedestrian level comfort. The presence of GI however in cities is very heterogeneous, and so is the case in the Brussels Capital Region (BCR) (Van de Voorde 2017). Urban GI implementations vary in -ratio, geometry, type (trees, bushes, green patches), and forest biodiversity.

Through the increasing accessibility to mobile environmental sensing technologies, pedestrian level environmental sensing campaigns are becoming progressively more popular contributing to the understanding of the strong spatial environmental gradients we are exposed to in cities, with potentially up to 8 times higher scalar concentrations within an urban block distance (Apte et al. 2017). However, while such studies are starting to report reductions in air temperature and improvements in urban comfort to be associated to the local presence of UGI, high spatial resolution direct sensing campaigns are yet limited, and further studies are required to characterize urban environmental gradients to high spatio-temporal resolutions as they relate to the local UGI characteristics. On the other hand, through the increasing access to Extended Reality (XR), eye tracking, and wearable biometric sensing technologies, the study of nature and urban scene preferences and their impact on psychological stress levels or overall comfort, are starting to be studied (Liszio and Masuch 2019; Lu et al. 2020). However, prior literature that has made use of XR, and biometric sensing technologies, have generally focused on the study of dichotomic immersion comparisons -such as green vs not green or urban vs rural-. Furthermore, amongst them, the studies that evaluated the restorative effect of nature for SLR for example, solely focused on laboratory - off-site experiments. However, these studies have already shown the benefits of combining XR and eye-tracking technologies, with biometric sensing and subjective participant feedback questionnaires, to evaluate the multi-domain perception and impact of the visualized scenes on the participant stress levels. The research methodologies to aim for a holistic study for the evaluation of thermal comfort and stress are yet to be further defined, however prior studies that have made use of the above-mentioned methodologies, have proved efficient to capture pedestrian level microclimate gradients at high spatial resolutions, and to combine biometric and eye tracking datasets with participant feedback questionnaires to reveal heat stress exposure indicators, or visual exploration patterns and heart rate variability associated to high stress levels.

2.3 Research project

This thesis is based on the working hypothesis that not only large scale, but also small to medium scale UGI that is integrated within the urban fabric, will modulate pedestrian level comfort (thermal comfort and stress particularly). Thus, it will be the objective of this research to identify the role

distinct GI characteristics (e.g., the ratio, geometry, type - trees, bushes, green patches, forest biodiversity) play in the enhancement of TC and SLR.

The second working hypothesis is that there will be significant correlations between SLR and PET. Thus, the objective will be to highlight the role environmental stressors, such as the exposure to increased temperatures, play in urban comfort, at physiological and psychological levels.

Designing an urban square to promote thermal comfort, emotional wellness, and reduce physiological stress levels is a challenging task given that pedestrian level physiological and psychological reactions are triggered by numerous environmental factors. The third working hypothesis of this research is that for the study of thermal comfort and stress, it is necessary to adopt a multi-domain (including visual, thermal and acoustic parameters) and pedestrian-centered research methodology. This research will not only include quantitative metrics, but also subjective measures to account for the participant's perception of environmental, physiological and psychological dimensions of urban comfort.

To our knowledge no comprehensive study focusing on urban comfort evaluation and particularly thermal comfort and stress has been developed to date where i) microclimate sensing, ii) high spatial resolution urban GI documentation, iii) biometric and eye-tracking measurements, and iii) subjective participant assessment, have been combined (see Fig.1). This research will build on the work developed through the one-year long Louvain4City grant obtained in 2021 "Brussels Street Greenery and Citizen Comfort Evaluation" by Prof. Llaguno-Munitxa (promoter) and developed in collaboration with Prof. Edwards (co-promoter), where virtual reality and biometric sensing based 'off-site' -or in laboratory- participant testing protocols were explored (see Fig. 2 & 3). This research will further develop the UGI quantification and participant testing strategies proposed for the Louvain4City grant, and will aim at the formulation of not only off-site but also on-site urban participant testing protocols.

With the ambition to test this working methodology, the research will start with a systematic review and city level UGI analysis, and will follow with the preparation of the experimental campaigns. The final year of the research will be focused to the data analysis and finalization of the thesis writing.

2.4 Work plan

The thesis will be organized in 4 main stages that will be developed during the 4 years of the proposed research. The research will adopt the Brussels Capital Region (BCR) as the case study.

1) Systematic Review (4 months):

A systematic review on urban GI for improved PET and SLR will be developed. This will form the first paper of the PhD. At the same time, we will apply for an ethics agreement for human participants testing.

2) UGI documentation: (8 months):

Remote sensing data to compute the Land Surface Temperature (LST) and the Normalized Difference Vegetation Index (NDVI), and UrBIS LiDAR aerial point cloud datasets will be used to quantify the UGI in BCR squares. For a selection of squares, a high spatial-resolution pedestrian level UGI characterization through segmentation analysis and semantic classification – using a LiDAR Faro-M70 scanner and an Insta-Pro 360° camera (available in LOCI-MiiL) - will be developed. The proposed GI quantification methodology will be published as a technical paper. Aside from the UGI quantification and characterization, the collected 360° videos and 3D scans,

will enable the projection of the recorded urban square scenes in a VR environment for the development of the off-site experimental campaign as per phase 4.1.

3) Experimental campaign preparation; (12 months)

The PET will be computed using a sensing-kit including: sound, temperature, relative-humidity, wind speed, wind direction, barometric pressure, and solar radiation sensors. Wearable sensors for Heart Rate Variability (HRV) and Skin Conductance (SC) will be used to identify the Stress Level Reduction (SLR) (Dalmeida and Masala 2021; Edwards et al. 2015). Through the XR headsets -VivePro HTC and Pupil core-, eye-tracking data will be collected - gaze, pupil diameter, blinks and head position-, and visual fixations and Areas of Interest (AoI) will be computed. These metrics will be used as indicators to identify UGI interest zones, and associate them with PET reduction and SLR. Participant feedback questionnaires such as the state Trait Anxiety Inventory (STAI) (Spielberger 1983), the Positive and Negative Affect Schedule (PANAS) (Watson et al. 1988), the Perceived Restorativeness Scale (PRS) (Hartig et al. 1997), and Connectedness to Nature Scale (CNS) (Navarro et al. 2017) will be included in the experimental campaign. The biometric and survey variables will be studied as the dependent variables and the root mean square of the successive differences and ANOVA for the repeated measures will be studied for the selected squares with distinct UGI characteristics. A methods paper describing the planning of the experimental campaigns will be published.

4) Experimental campaigns:
Following the approval of the ethics committee, the experimental campaigns will be initiated by by an intensive participant recruiting process.

4.1. Experimental campaign Off-site (8 months):
The participants will be tested in the XR OpenHub in LLN using biometric sensors for HRV and SC, and feedback questionnaires to report the SLR. Using the 360° videos and scans captured during phase 2, the urban scenes will be visualized in a VR headset - VivePro HTC, and through its implemented eye-tracking technologies, the gaze behavior will be analyzed to identify the AoI and fixations (Edwards et al. 2003). The sounds of the recorded urban scenes will also be preserved, and situations where very intrusive objects dominate the urban scene will be avoided. The questionnaires will need to be filled in prior, during and after the experiment. The off-site protocol will enable a controlled environment to identify the percentage of time spent in the UGI AoI and fixations, and their relation with SLR. A paper describing the off-site experimental protocol and the research findings will be prepared.

4.2. Experimental campaign On-site (8 months):
The recruited participants will be transported to the selected squares in BCR (selection of ~12 squares made in phase 2), where they will carry a wearable sensing kit including the microclimate and biometric sensors (see list in 3) to measure the PET reduction and SLR associated to the quantified UGI characteristics (as per phase 2). The Pupil AR headset will be used to study the on-site UGI fixations and AoI. On-site pedestrian surveys will also be filled in. Through regression analysis, the UGI metrics and PET reduction, PET and SLR, and off-site and on-site measurements will be studied. The experimental campaign will be developed two times per day for all squares. Days with clear weather conditions, and with low urban occupancy will be selected for all studied sites. A paper describing the on-site experimental protocol and the research findings will be prepared.

Finally, in the last 8 months of thesis, the final data analysis protocols, and the comparison between the tested participant testing approaches will be evaluated. The papers initiated during the phases 4.1 and 4.2 will be concluded. The last months of the research will be dedicated to the thesis formatting and deposition.

Bibliography

Apte JS, Messier KP, Gani S, et al (2017) High-Resolution Air Pollution Mapping with Google Street View Cars: Exploiting Big Data. Environ Sci Technol 51:6999–7008. https://doi.org/10.1021/acs.est.7b00891

Cole MA, Ozgen C, Strobl E (2020) Air Pollution Exposure and Covid-19 in Dutch Municipalities. Environ Resour Econ 76:581–610. https://doi.org/10.1007/s10640-020-00491-4

Dalmeida KM, Masala GL (2021) HRV Features as Viable Physiological Markers for Stress Detection Using Wearable Devices. Sensors 21:2873. https://doi.org/10.3390/s21082873

Edwards MG, Humphreys GW, Castiello U (2003) Motor facilitation following action observation: A behavioural study in prehensile action. Brain Cogn 53:495–502

Edwards MG, Jiang D, Mullins P, Callow N (2015) The neural substrates for the different modalities of movement imagery. Brain Cogn 97:22–31

Founda D, Santamouris M (2017) Synergies between Urban Heat Island and Heat Waves in Athens (Greece), during an extremely hot summer (2012). Sci Rep 7:1–11

Hartig T, Korpela K, Evans GW, Gärling T (1997) A measure of restorative quality in environments. Scand Hous Plan Res 14:175–194

Imran HM, Kala J, Ng AWM, Muthukumaran S (2019) Effectiveness of vegetated patches as Green Infrastructure in mitigating Urban Heat Island effects during a heatwave event in the city of Melbourne. Weather Clim Extrem 25:100217

Jayasooriya VM, Ng AWM, Muthukumaran S, Perera BJC (2017) Green infrastructure practices for improvement of urban air quality. Urban For Urban Green 21:34–47

Johnson S, Ross Z, Kheirbek I, Ito K (2020) Characterization of intra-urban spatial variation in observed summer ambient temperature from the New York City Community Air Survey. Urban Clim 31:100583. https://doi.org/10.1016/j.uclim.2020.100583

Liszio S, Masuch M (2019) Interactive immersive virtual environments cause relaxation and enhance resistance to acute stress. Annu Rev Cyberther Telemed 17:65–71

Llaguno-Munitxa M, Bou-Zeid E (2020) The environmental neighborhoods of cities and their spatial extent. Environ Res Lett 15:074034

Lu S, Sanchez Perdomo YP, Jiang X, Zheng B (2020) Integrating eye-tracking to augmented reality system for surgical training. J Med Syst 44:1–7

Manoli G, Fatichi S, Schläpfer M, et al (2019) Magnitude of urban heat islands largely explained by climate and population. Nature 573:55–60

Navarro O, Olivos P, Fleury-Bahi G (2017) “Connectedness to nature scale”: Validity and reliability in the French context. Front Psychol 8:2180

Orchard S, Schiel DR (2021) Enabling nature-based solutions for climate change on a peri-urban sandspit in Christchurch, New Zealand. Reg Environ Change 21:1–18

Pigliautile I, Pisello AL (2018) A new wearable monitoring system for investigating pedestrians’ environmental conditions: Development of the experimental tool and start-up findings. Sci Total Environ 630:690–706

Piselli C, Castaldo VL, Pigliautile I, et al (2018) Outdoor comfort conditions in urban areas: On citizens’ perspective about microclimate mitigation of urban transit areas. Sustain Cities Soc 39:16–36

Sharifi A, Khavarian-Garmsir AR (2020) The COVID-19 pandemic: Impacts on cities and major lessons for urban planning, design, and management. Sci Total Environ 749:142391. https://doi.org/10.1016/j.scitotenv.2020.142391

Sowińska-Świerkosz B, Wójcik-Madej J, Michalik-Śnieżek M (2021) An Assessment of the Ecological Landscape Quality (ELQ) of Nature-Based Solutions (NBS) Based on Existing Elements of Green and Blue Infrastructure (GBI). Sustainability 13:11674

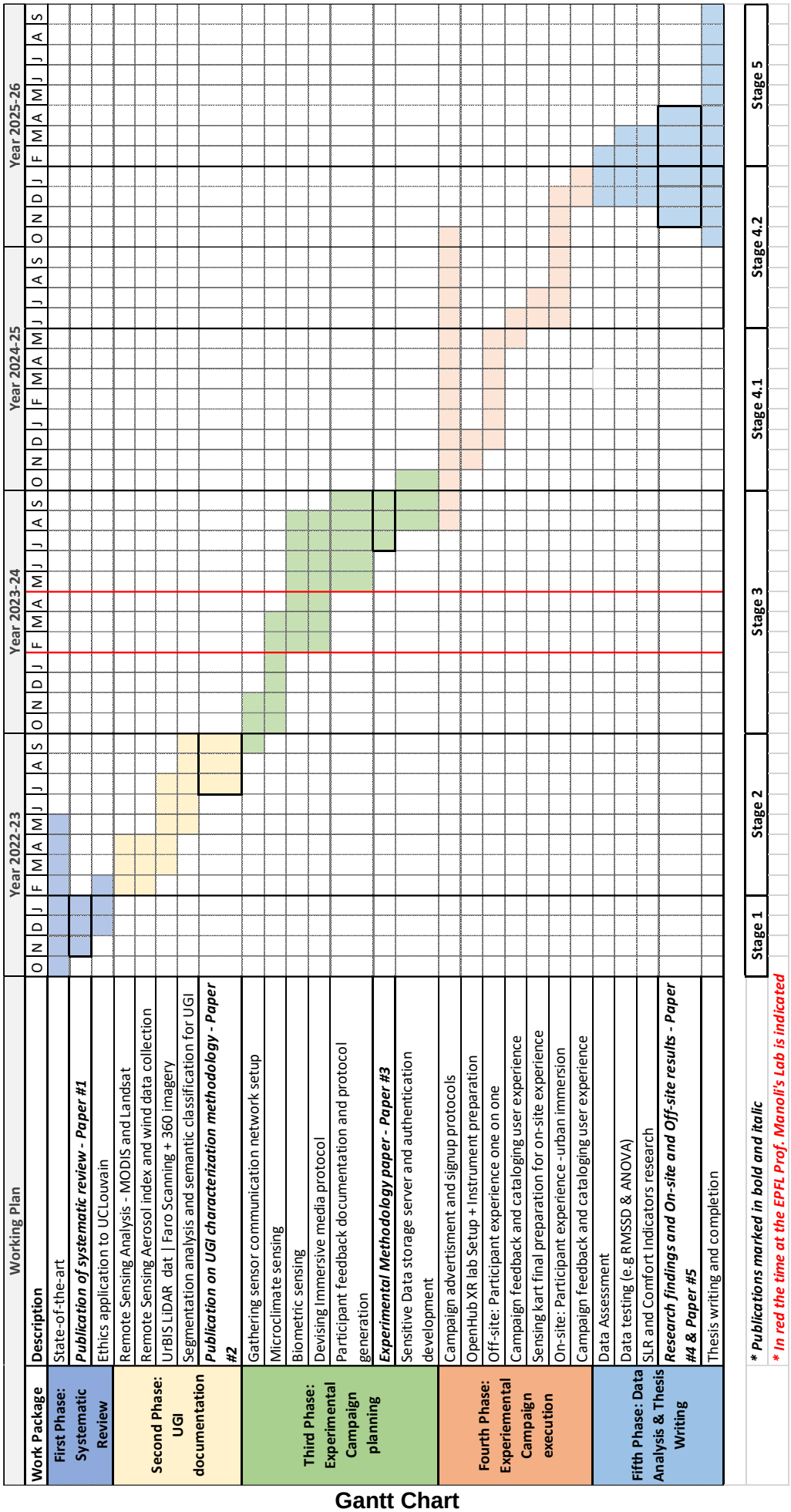
Spielberger CD (1983) State-trait anxiety inventory for adults

Tawatsupa B, Yiengprugsawan V, Kjellstrom T, et al (2012) Heat stress, health and well-being: findings from a large national cohort of Thai adults. BMJ Open 2:e001396. https://doi.org/10.1136/bmjopen-2012-001396

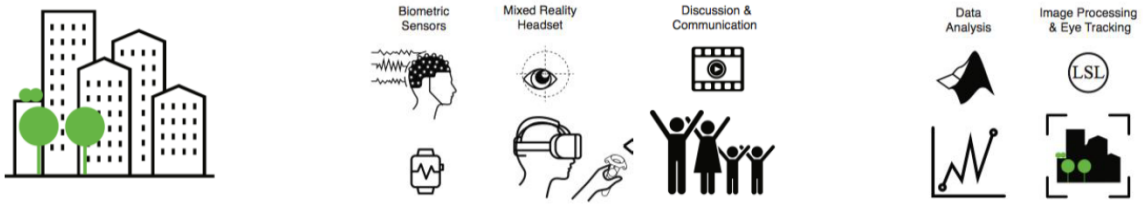
Van de Voorde T (2017) Spatially explicit urban green indicators for characterizing vegetation cover and public green space proximity: a case study on Brussels, Belgium. Int J Digit Earth 10:798–813. https://doi.org/10.1080/17538947.2016.1252434

Villadiego K, Dabat M-A (2014) Outdoor Thermal Comfort in a Hot and Humid Climate of Colombia: A Field Study in Barranquilla. Build Environ 75:. https://doi.org/10.1016/j.buildenv.2014.01.017

Watson D, Clark LA, Tellegen A (1988) Development and validation of brief measures of positive and negative affect: the PANAS scales. J Pers Soc Psychol 54:1063


Work Package
Description
Working Plan
Year 2022-23
Year 2023-24
Year 2024-25
Year 2025-26
O N D J F M A M J J A S O N D J F M A M J J A S O N D J F M A M J J A S
First Phase: Systematic Review
State-of-the-art
Publication of systematic review - Paper #1
Ethics application to UCLouvain
Second Phase: UGI documentation
Remote Sensing Analysis - MODIS and Landsat
Remote Sensing Aerosol index and wind data collection
UrBIS LiDAR dat | Faro Scanning + 360 imagery
Segmentation analysis and semantic classification for UGI
Publication on UGI characterization methodology - Paper #2
Third Phase: Experimental Campaign planning
Gathering sensor communication network setup
Microclimate sensing
Biometric sensing
Devising Immersive media protocol
Participant feedback documentation and protocol generation
Experimental Methodology paper - Paper #3
Sensitive Data storage server and authentication development
Fourth Phase: Experiemental Campaign execution
Campaign advertisment and signup protocols
OpenHub XR lab Setup + Instrument preparation
Off-site: Participant experience one on one
Campaign feedback and cataloging user experience
Sensing kart final preparation for on-site experience
On-site: Participant experience -urban immersion
Campaign feedback and cataloging user experience
Fifth Phase: Data Analysis & Thesis Writing
Data Assessment
Data testing (e.g RMSSD & ANOVA)
SLR and Comfort Indicators research
Research findings and On-site and Off-site results - Paper #4 & Paper #5
Thesis writing and completion
Stage 1
Stage 2
Stage 3
Stage 4.1
Stage 4.2
Stage 5
* Publications marked in bold and italic
* In red the time at the EPFL Prof. Manoli's Lab is indicated

Gantt Chart



- 3D scanning (Point cloud & video)
- Microclimate survey (direct sensing)

- Biometric sensing
- Immersive media
- Questionnaires

- Participant data analysis
- Urban GI and environmental data analysis

Urban GI and microclimate data

Participant data, on-site and off-site immersion

Data analysis

Figure 1: Research workflow diagram

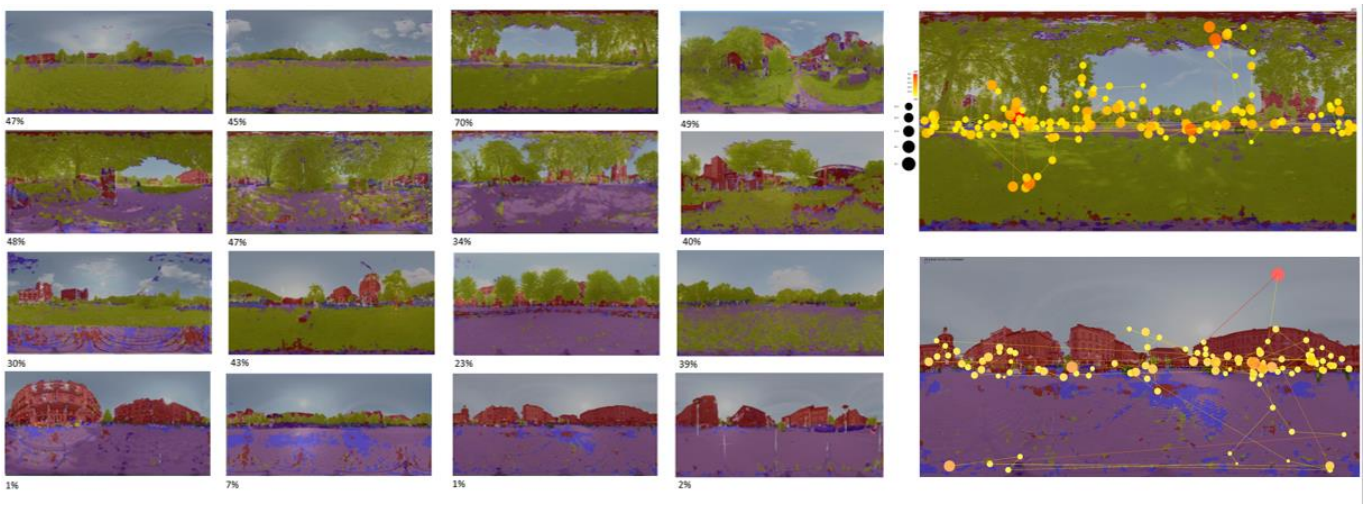


Figure 2: (left) video & point cloud semantic classification of 16 sites in BCR with distinct green index ratios for characterizing green infrastructure (right) eye tracking data visualization (AOI & fixations)





- State-Trait Anxiety Inventory (STAI)
- Positive And Negative Affect Schedule (PANAS)
- Perceived Restorativeness Scale (PRS)
- Connectedness to Nature Scale (CNS)

Continuous measurement of Heart Rate Variability (and SCL).

Figure 3: (left) Participant recruitment announcement launched in December 2021. (right) participant testing within the openHub in Louvain-La Neuve

3. COMMENTS ON CHANGES MADE IN THE RESEARCH PROJECT IN CASE OF RESUBMISSION (OPTIONAL)

NA

4. POTENTIAL INTERDISCIPLINARY APPROACH OF THE RESEARCH PROJECT (OPTIONAL)

The proposed research focuses on the study of the role urban green infrastructure characteristics play on urban comfort - stress level reduction and thermal comfort in particular. This research requires the development of quantitative and qualitative, interdisciplinary research methodologies that combine: urban science, environmental science, experimental psychology and cognitive neuroscience perspectives.

My research experience is focused on the study of environmental sciences (focusing on thermal comfort and environmental sustainability), resource allocation, cloud computing and landscape planning, classification and prediction maps for forecasted years, data fusion research and multi-criteria decision analysis. Given the interdisciplinarity of the research proposal, my research expertise will be complemented with the environmental and urban science background of promoter Prof. Llaguno-Munitxa from UCLouvain - Louvain Landscape, Architecture and Built (LAB) institute, and with the expertise on experimental psychology and cognitive neuroscience provided by the co-promoter, Prof Edwards president of the UCLouvain Psychological Sciences Research Institute (IPSY). An international collaboration with Prof. Manoli from the University College London - Department of Civil, Environmental & Geomatic Engineering has also been established. Prof. Manoli is specialized in environmental sciences and urban forestry.

My research background will be complemented by that of Prof. Llaguno-Munitxa and the research group, Architecture et Climat group of LAB at UCLouvain. The Architecture et Climat group fundamental and applied research interests include, amongst others, the study of urban science, climate and health as well as the study of environmental quality and human comfort factors. The urban analysis workflows for the quantification and characterization of the urban green infrastructure, and the environmental sensing campaign preparation (sensor testing and calibration), will be developed within the Architecture et Climat lab and under the supervision of Prof. Llaguno-Munitxa. The expertise of Prof. Manoli and his collaborators from the Urban Ecohydrology Lab at the University College London, will also be instrumental to the development of the sensing protocol. Prof Manoli will be joining the School of Architecture, Civil and Environmental Engineering at EPFL as professor in September 2022, where he will be hosting me in his lab during a stay of 90 days during the phase 3 of the research.

The preparation of the biometric sensing protocol and participant questionnaire methodology and testing will be developed in collaboration with (co-promoter) Prof. Edwards and his collaborators from the Psy-NAPS lab from IPSY- UCLouvain, (associated with the Institute of Neuroscience and Louvain Bionics). The contribution of Prof. Edwards will focus on the guidance of the VR experimental approaches, the biometric sensing protocol and the use of participant questionnaire methodology, as well as providing support equipment from his Psy-NAPS group.

The research that I will be working on, will benefit from the interdisciplinary project developed as part of the grant Louvain4City 2021 “Brussels Street Greenery and Citizen Comfort Evaluation” PI- Asst Prof. Llaguno-Munitxa. Within UCLouvain, this work was performed in collaboration with Prof. Edwards (president of IPSY at UCLouvain) and the Media Innovation & Intelligibility Lab (MiiL) through the contribution of Prof. Sebastien Nahon, as well as through the collaboration with the OpenHub XR team led by Dr. Sonneville. Aside from the internal UCLouvain collaborations, the Louvain4City grant initiated an international collaboration I would like to build on with Prof. Tzyy Ping, co-Director of the Center for Advanced Neurological Engineering, Institute for Neural Computation (INC) and Institute of Engineering in Medicine (IEM), in the University of California, San Diego. Furthermore, a collaboration with distinguished Prof. Eduardo Macagno from the department of Developmental Biology at UCSanDiego and specialized in the novel interdisciplinary area: Architecture and Neuroscience was established. Prof. Eduardo Macagno is also co-Director of the Academy of Neuroscience for Architecture.

5. DESCRIPTION OF THE WORK ENVIRONMENT

The promoter Prof. Llaguno-Munitxa is a professor at UCLouvain where she is a member of the faculty LOCI (Architecture, Architectural Engineering, Urbanism), and the Louvain Landscape, Architecture and Built environment (LAB) research institute (created in January 2021 to sustain, coordinate, nurture and manage all research activities at UCLouvain in the fields of architecture, architectural engineering, urban and landscape design, and territorial development). Within LAB, the research group Prof. Llaguno-Munitxa is part of, the Architecture et Climat group - founded in 1980, is one of the most recognized and longest-standing entities within UCLouvain.

As PhD candidate, I will be part of the Louvain research Institute LAB and within it, the Architecture et Climat research group. The group focuses its fundamental and applied research interests on four interrelated areas of 1) urban science, climate and health; 2) life cycle performance and parametric modelling at urban and building scale; 3) building performance simulation and energy flows predictions 4) environmental quality and human factors. The research of Prof Maider Llaguno-Munitxa is focused on the first pilar of research within the group, that is the study of urban science, comfort and health. The research methodologies she uses in her work include direct sensing, environmental modeling, and data-driven approaches. Working alongside several national and international partners, the architecture et climat group has developed significant expertise in the development of energy performance simulation modelling and analysis of dynamic climate-based phenomena including the thermal comfort and ventilation. The group regularly publishes papers in peer-reviewed international journals and is invited at national and international conferences. Within Architecture et Climat, over 30 PhD theses have been completed since its creation, including several FNRS-funded scholarships. Furthermore, through the collaboration with the co-promoter Prof. Edwards, and his lab members at Psy-NAPS, I will engage with biometric sensing technologies for the heart rate variability, skin conductance as well as the potential usage of Electroencephalography. This collaboration will also be critical for the analysis and rationalization of the collected datasets, and development of the participant survey protocols. My stay during 3 months at the EPFL lab of Prof. Manoli (currently at the University College London) will enable the further testing and evaluation of the microclimate sensing protocol and the computation of the Perceived Equivalent Temperature (PET).

Selected list of publications of the promoter and collaborators (related to this project):

Llaguno-Munitxa M., Bou-Zeid E. Sensing the Environmental Neighborhoods. (2021). Proceedings of the DigitalFUTURES. CDRF 2020. Springer, Singapore.

Llaguno-Munitxa M, Bou-Zeid E. (2020). The environmental neighbourhoods of cities and their spatial extent. Environmental Research Letters, 15.

laguno-Munitxa M, Bou-Zeid E, Hultmark M. (2017). The influence of building geometry on street canyon air flow: validation of large eddy simulations against wind tunnel experiments. J. Wind Eng. Ind. Aerod., 165.

Callow, N, Edwards, M. G, Jones, A. L., Hardy, L., & Connell, S. (2019). Action dual tasks reveal differential effects of visual imagery perspectives on motor performance. Quarterly Journal of Experimental Psychology, 72(6).

Grade S, Pesenti M and Edwards MG. (2015). Evidence for the embodiment of space perception: concurrent hand but not arm action moderates reachability and egocentric distance perception. Front. Psychol. 6:862.

Manoli, G, Fatichi, S, Schläpfer, M. *et al.* (2019). Magnitude of urban heat islands largely explained by climate and population. Nature 573**.**

Meili, N, Manoli, G, Burlando, P, Carmeliet, J, Chow, W, Coutts, A, Roth, M, Velasco, E, Vivoni, E, Fatichi, S (2021). Tree effects on urban microclimate: Diurnal, seasonal, and climatic temperature differences explained by separating radiation, evapotranspiration, and roughness effects. Urban Forestry & Urban Greening, 58.

6. SUMMARY OF THE MASTER'S THESIS OR EQUIVALENT

Urban Heat Environment is subjective to various phenological & anthropogenic changes throughout the day. However, satellites have always faced gain and loss bias towards prioritizing spatial resolution vs temporal resolution, which has accounted for various time gaps between consecutive observations. Gaining higher spatial resolution would require satellite to be placed in near-earth orbit, which takes longer time for instrument to circle back around the world to initial position for a secondary observation. Additionally, detailed observations require higher radiometric capability, which requires more ground controls to receive data, which makes the entire enterprise expensive. And near-earth satellites have historically used extensive fuel to maintain orbital position, resulting in a lower lifespan. Such problems prevent satellite data to give a complete overview of the Earth phenomenon, and to compensate for such gaps in satellite sensing, ground sensors were developed to sense any desired parameter and store locally or transmit data under private radio frequencies. Such sensing is referred to as hyper-local sensing and accounts for sensing phenomena irrespective of daylight and can help fill voids in satellite sensing.

But as the sensors act as a proxy of satellite sensing, the location of placing the sensor has been of great debate especially since the exact location to place sensor is subjective to individual observers. A standard approach, if it were to be developed, would require a holistic observation of the entire region to pinpoint location based on environment.

The use of this study will provide an aggregated framework for carrying out sensor placement, which will be focused on the phenomenon being perceived and unbiased from human perception and will be more efficient technologically.

Spatiotemporal data fusion was useful in prioritizing the grid cells that have more influence to the urban heat gains, and sensor placement strategy which was based on these priorities were carried out. Further, the varieties of existing LCZ within grids were considered and estimate of the overall sensors required for monitoring these features were mathematically established. And further the authorities that have a set budget for purchasing the sensors, can use the proposed method for understanding the minimum number of sensors needed for a city-wide sensing and along with locations that can be prioritized.

This prioritization scheme was purely dependent on the measured heat intensities over surfaces and the network source (LoRa Gateway) placement was based on the maximum coverage potential under the terrain considerations of study area, and thereby optimization of network sources was also carried out. These network serves have to be placed in secure locations, and therefore the government buildings and public pre-existing infrastructure was mapped with OSM data, where sufficient range can be achieved for end sensor placement.

Finally with the established variability at the end device, gateway, terrain consideration and public infrastructure availability, the sensor placement map was proposed which would be optimal and unbiased towards the Local Climate Zones being studied, and would be based on the phenomenon of Urban Heat being sensed.

7. ADDITIONAL COMMENTS (OPTIONAL)

NA